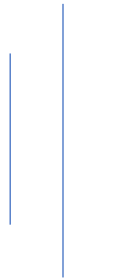


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A REPORT OF STRUCTURAL ANALYSIS I



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**OBJECTIVE 1: TO DETERMINE DEFLECTION AT MIDSPAN OF SIMPLY SUPPORTED BEAM
SUBJECTED TO POINT LOAD AT MID SPAN.**

INSTUMENTS REQUIRED:

1. Simply supported beam
2. Point load (Weights)
3. Measuring tape/ruler
4. Vernier caliper
5. Supports for beam
6. Dial gauge

THEORY: When a simply supported beam is subjected to a point load applied at its midspan, the beam undergoes deflection due to the applied load. The deflection is caused by the bending of the beam, as the load induces a moment that causes the beam to deform.

In simply supported beam, supports are provided at both ends of a beam. The theoretical deflection at mid span in a beam due to point load is given by:

$$\Delta = PL^3/48EI$$

Where, E=Modulus of elasticity of beam

I= Moment of inertia of beam

$$=bd^3/12$$

Whereas practical deformation can be obtained with the help of dial gauge readings i.e. the practical deformation= final gauge reading- initial gauge reading

PROCEDURE:

1. Length of the beam was measured using ruler.
2. Width and depth of beam was measured using Vernier caliper.
3. Dial gauge was fitted at midspan of beam.
4. Initial reading of dial gauge before the weight was placed on the hanger placed at midspan was noted.
5. Then weight was suspended on the hanger at midspan and corresponding readings on dial gauge were noted.
6. Theoretically deflection at midspan corresponding to each weight was calculated using above formula.
7. Deflection at midspan of beam was also calculated by taking the difference between the readings taken.
8. The graph of theoretical deflection and experimental deflection was drawn.

9. Both the graph obtained were compared to each other and conclusion was drawn.

OBSERVATION AND OBSERVATION TABLE:

TABLE 1:

Meaning	Symbol	Value	Unit
Length	L	700	mm
Width	B	23.73	mm
Depth	D	5.14	mm
Elasticity	E	2.01×10^5	N/mm ²
MOI	I	268.5380613	mm ⁴

TABLE 2:

Load in N		Dial Gauge Reading(mm)		Practical deflection(mm)	Theoretical deflection(mm)	
S.N.	W	Initial(x)	Final(y)	$D_1=y-x$	$D_2=PL^3/48EI$	ERROR=(d_2-d_1)/ $d_2\%$
1	5.47398	4.36	4.84	0.48	0.724693194	33.73%
2	11.2815	4.36	5.48	1.12	1.49354332	25.01%
3	17.14788	4.36	6.34	1.98	2.270185846	12.78%
4	23.17122	4.36	7.09	2.73	3.067608106	11.01%
5	28.10565	4.36	7.73	3.37	3.720870967	9.43%

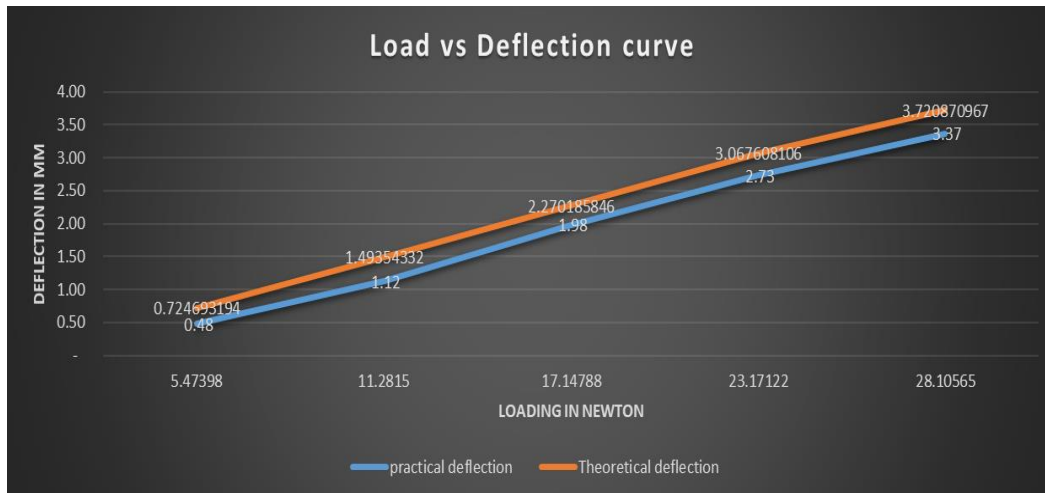


FIG: Graphical comparison betn theoretical and practical deformation

Conclusion:

The graph between the load and displacement at midspan drawn theoretically and practically , and percentage error was calculated.

OBJECTIVE 2: TO DETERMINE ILD FOR A SIMPLY SUPPORTED BEAM DUE TO MOVING LOAD.**INSTRUMENTS REQUIRED:**

1. Weight
2. Hanger
3. Simply Supported Beam
4. Ruler
5. Dial gauge

THEORY: An Influence Line Diagram (ILD) is a graphical representation that illustrates how a particular reaction (force or moment) at a specific point on a structure varies as a unit load moves along the length of the structure. In this case, we are interested in determining the ILD for a simply supported beam subjected to a moving load

Theoretical Deflection at midspan when load acting is not at center is given by:

$$\Delta = Pb(3l^2 - 4b^2)/48EI, a > b$$

$$\Delta = Pa(3l^2 - 4a^2)/48EI, a < b$$

PROCEDURE:

1. The length of given beam was measured.
2. Dial gauge was fitted at midspan of beam.
3. Initial dial gauge reading was noted.
4. Load of constant magnitude was moved at constant distances from a support in the given beam.
5. Dial gauge reading for different position of load was taken.
6. Finally, the displacement was calculated by taking the difference between the initial and final reading.
7. The graph was plotted between load position and corresponding displacement for both experimental and theoretical values.

OBSERVATION AND OBSERVATION TABLE:

TABLE 1:

Meaning	Symbol	Value	Unit
Length	L	700	mm
Width	B	23.73	mm
Depth	D	5.14	mm
Elasticity	E	2.01×10^5	N/mm ²
MOI	I	268.5380613	mm ⁴

TABLE 2:

13	11.01663	650	50	4.36	4.58	0.22	0.310355659	29.11%
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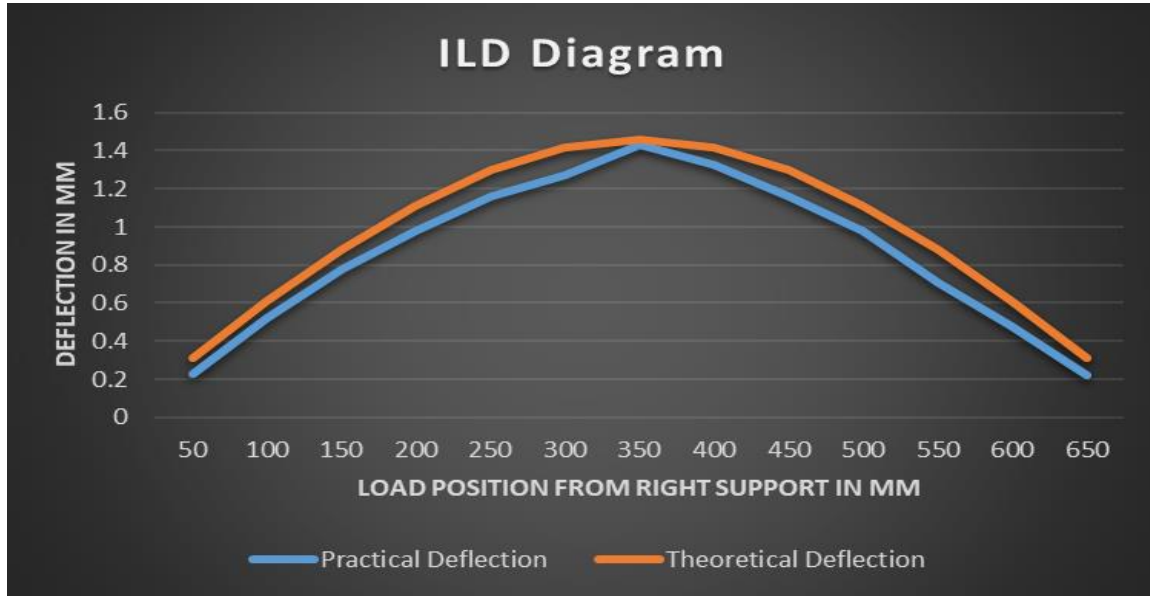


FIG: Graphical representation of comparison betn theoretical and practical deformation at midspan due to moving load

CONCLUSION:

Hence ILD for simply supported beam at midspan was determined and drawn as above.

OBJECTIVE 3: TO DETERMINE THE ILD FOR HORIZONTAL THRUST OF THREE HINGED ARCH.

APPARTUS REQUIRED:

1. Support frame
2. Three hinged arch assembly
3. A simple support
4. A roller support

THEORY:

There are mainly three types of arches that are commonly used in practice:

- a) three hinged arch
- b) two-hinged arch and
- c) fixed arch

The three-hinged arch is statically determinate structure, and its reactions / internal forces are evaluated by static equations of equilibrium. Two-hinged arch and fixed-fixed arch are statically indeterminate structures. The indeterminate reactions are

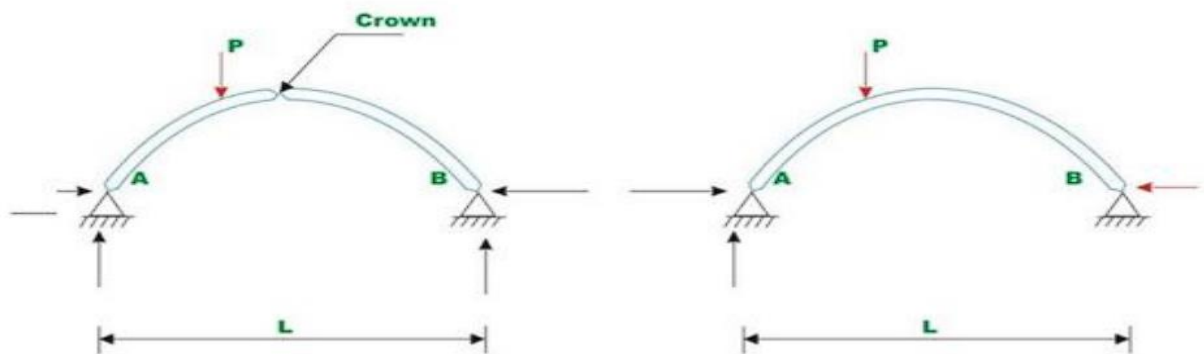
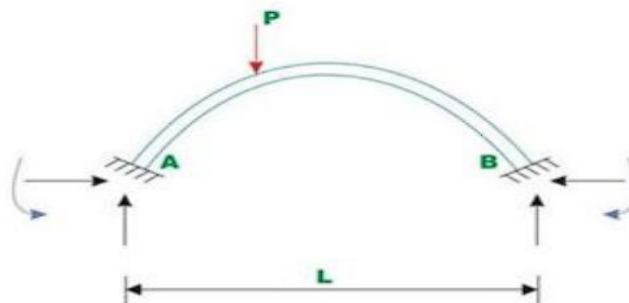


Fig a: Three Hinged Arch

Fig b: Two Hinged Arch



determined by the method of least work or by the flexibility matrix method. In this experiment three hinged arch is discussed.

To find the theoretical values of horizontal thrust,

$$H_A = \frac{Wk}{2h}$$

Where,

W is load

H is height

K is the distance of the load from the pinned support

PROCEDURE:

1. The model of the three-hinged arch was assembled in the laboratory.
2. The model was ensured securely mounted on its supports.
3. Lubricating oil was used at the roller end of the arch by placing load on the hanger for horizontal thrust until the best equilibrium conditions were obtained.
4. A few loads were placed on the arch in any chosen position. These were balanced by placing additional weights on the hanger for horizontal thrust. The additional weights added on the thrust hanger gave the experimental value of the horizontal thrust.
5. A load of 2kg was placed in turn on each hanger one by one and required balancing weight required on the thrust hanger was found.
6. The ordinate representing $\frac{1}{2}$ of the balancing weights on the load positions were plotted as base.
7. This gave the influence line diagram for horizontal thrust.

CALCULATION:

When load is placed at center, $k = 50\text{cm}$, then

$$\text{For load of } 5\text{N}, H = 5 * \frac{50}{2*25} = 5\text{N}$$

$$\text{For load of } 10\text{N}, H = 10 * \frac{50}{2*25} = 10\text{N}$$

$$\text{For load of } 15\text{N}, H = 15 * \frac{50}{2*25} = 15\text{N}$$

Due to the fault of instrument, we could not obtain experimental value and unable to compare the theoretical and practical value of horizontal thrust.

CONCLUSION: Study about the three-hinged arch is done and the theoretical value was obtained.

OBJECTIVE 4: ANALYSIS OF THE DEFLECTION AT JOINTS OF TRUSS

APPARATUS REQUIRED: 1. Truss apparatus

2. Weights

3. Hanger

4. Dial Gauge

5. Scale and vernier caliper

THEORY:

The deflection of a node of a truss under a given loading is determined by:

$$\delta = \Sigma (F \cdot U \cdot L / AE)$$

Where,

δ = deflection at the node point.

F = Force in any member under the given loading.

U = Force in any member under a unit load applied at the point where the deflection is required. The unit load acts when the loading on the truss has been removed and acts in the same direction in which the deflection is required.

L = Length of any member.

A = Cross sectional area of any member.

E = Young's modulus of elasticity of the material of the member. Here, (L/AE) is the property of the member, which is equal to its extension per unit load. It may be determined by suspending a weight from it and noting the extension.



OBSERVATION:

The truss structure was observed and we were unable to take data as truss structure was not functioning properly.

RESULT:

The idea about truss behavior and deflection in truss due to different loading was taken theoretically.

OBJECTIVE 5: ANALYSIS OF COMPONENTS OF SUSPENSION BRIDGE.

INSTRUMENT REQUIRED : A suspension bridge apparatus

TERMS:

1. **Main cables:** Simulated the main load-bearing cables.
2. **Vertical suspenders:** Connected the main cables to the bridge deck.
3. **Bridge deck:** The platform upon which loads were applied.
4. **Load application system:** Consisted of weights and a mechanism to apply controlled loads to the bridge deck.
5. **Deflection measurement system**



Fig: Suspension Bridge Apparatus

DISCUSSIONS:

1. As the load was applied, the main cables experienced tension, and the vertical suspenders carried this tension from the main cables to the bridge deck.
2. There were unable to take any data due to fault in truss structure.

RESULT:

The basic idea about analysis of truss and deflection in it due to different loading was obtained.